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Report: Analysis of Hospital Management Systems (HMS)

This research paper focuses on a technical and managerial overview of a Hospital Management System (Hospital Information System) – the key software in the modern health care sector. A hospital management system (HMS) is considered as a vital tool for storing and managing the data of a health entity. The paper explains the purpose of the HMS and its key users – doctors, nurses, and patients. The main body of the report analyzes functional and non-functional requirements for the software, stakeholders, and other processes such as risks and quality attributes. The work also reviews the use of artificial intelligence in hospital management systems and its risks. Finally, the article concludes by emphasizing the importance of this technology for the health care industry.

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1. Purpose of the Hospital Management System

The central goal of a Hospital Management System is to optimize the hospital's functioning through the digitalization of the process. Thus, the key purposes of the software are listed below:

- **Making the operations of the hospital smoother:** The system allows us to automate numerous official duties such as the record creation for patients, the bills, and receipts' generation, and other administrative tasks. As a result, employees reduce the time spent on these duties, making the organization of the hospital easier.
- **Improving the quality of the patient's treatment:** For instance, the doctor gets access to the patient's medical history and all previous diagnostic results. To diagnose patients, physicians gather information from the body as well as the mind and behavior. the diagnosis rationally. Another benefit is the right use of the hospital's resources: beds, equipment, personnel, and materials.

2. Users of the System

A software application has users who interact with it. According to the work, a Hospital Management System has several types of end-users:

- **Doctors (Physicians and Specialists):** This category of workers uses the application to input patient data, prescribe drugs, order laboratory and instrumental studies, view results, and perform other clinical duties.
- **Nurses and Clinical Support Staff:** They interact with the application to observe patients' conditions and manage dispensary procedures such as medication scheduling and transportation from one hospital department to another.
- **Administrative and Reception Staff:** This division utilizes the application to make phone calls and messages to patients, input data into the system, maintain the master patient index, make appointments, respond to inquiries, and perform other clerical duties.
- **Patients:** With the development of the concept, patients are not only recipients of the organization's services but also its users. The application offers these individuals the following features: making an appointment, accessing medical records, downloading reports, and paying for the services.
- **Ancillary Department Staff (Pharmacy, Lab, Radiology):** This division employs the system to manage pharmaceutical stocks, provide the clinicians with the dose instructions, and store laboratory tests in patients' electronic health records.

3. Stakeholders

There are also some stakeholders who are interested in the functioning of the software. Stakeholders are people, groups, or organizations that use the software, depend on its information, provide services to the organization, or are affected by the software. In a Hospital Management System (HMS/HIS), different stakeholders have different responsibilities and interests. For instance:

- **Hospital Executives and Management:** This stakeholder will use HMS/HIS reports to control the functioning of the hospital and to strategize its growth. The management will

use the generated reports to monitor the progress of the hospital, control activities in the departments, analyze the hospital's finances, and make informed decisions

- **Insurers:** As the payers for the services rendered, this stakeholder will be interested in the quality and cost of services offered. Insurance companies can use this information to verify treatments, services, and suppliers of the hospital. Thus, when making payments, they will rely on this information to ensure that nothing is claimed that should not be covered by insurance.
- **Suppliers:** This stakeholder's relationship with the hospital is vital to ensure it has machines, equipment, and medicines on a continuous basis. Suppliers such as manufacturers of medicine, medical equipment, laboratory equipment, and surgical materials will depend on the software to provide the items ordered.
- **Government and Regulators:** Ensuring that the hospitals they regulate meet the standards is the primary role of this stakeholder. The agencies may set standards for healthcare, including privacy, safety, health, and hospital administration, and ensure they are implemented
- **IT Service Providers and Developers:** This stakeholder will have an interest in the software who will maintain and protect the system. As developers, this stakeholder will be involved in developing and testing techniques that can add value to the software. It will also ensure patients' records are secure and protected from unauthorized access

4.Functional Requirements

Functional requirements are system features that should be in a product to meet users' expectations. There are five functional requirements for a hospital management system:

- **Patient Registration and Unique Identification:** The first step to receiving hospital services is the patient's registration. The application should collect the individual's data (demographic, clinical) and assign a patient ID, which will be used to store this person's medical and financial records.
- **Appointment and Schedule Management:** The system should offer an easy-to-use schedule management function. It should enable users to set appointments by considering patients' needs and the availability of doctors and hospital resources (e.g., scan rooms).
- **Electronic Medical Record (EMR):** The application should store patients' clinical histories and offer this data to clinicians in real time. Moreover, the system should feature a complete range of previous treatments (e.g., surgeries, vaccination).
- **Billing:** The program should calculate the cost of services (e.g., consultations, medicines) and assign this information to patients' accounts. It should also offer various payment options (cash, cards, and online transfers).
- **Inventory and Pharmacy Management:** The software should keep track of the hospital's stock of medical supplies and pharmaceuticals. It should notify the inventory department when it is time to reorder (based on the set stock level). Moreover, the application should help to register patients' medication doses to ensure their safety.

5.Non-Functional Requirements

Non-functional requirements define product qualities and additional constraints.

- **Security and Data Protection:** Given that the application stores sensitive information, it should offer protection. The system should employ various safety measures (e.g., two-factor authentication, role-based access, and data encryption).
- **Usability and Interface:** The software's interface should be intuitive and user-friendly. This will ensure that the end-users will spend less time studying the application.
- **Performance and Response Time:** The system should respond quickly to end-users. Otherwise, its usage will be problematic for the employees.
- **High Availability and Reliability:** The application should be available 24/7 since it is a vital tool for the hospital's functioning. It should possess mechanisms that ensure that no one will be deprived of services in case of a malfunction.
- **Maintainability:** The system should be easy to maintain, update, and modify. Developers should be able to fix errors, add new features, and make changes without affecting the rest of the system.

6. System Risk:

The risk system of hospital system is:

- **System Failure:** A hospital system may stop working because of hardware, software, or network problems. When this happens, doctors and nurses may not be able to access patient records. It can delay appointments, diagnosis, and treatment. This can affect the quality and safety of patient care.
- **Data Loss** Important patient information may be lost because of system errors or technical problems. This information can include medical history, test results, and prescriptions. Without proper backups, recovering the lost data can be difficult. Data loss can cause delays and mistakes in patient treatment.
- **Cybersecurity Attacks:** Hackers may attack hospital systems to steal or damage patient information. Patient data such as medical records and personal details can be exposed. A cyberattack can also stop hospital services from working normally. Strong passwords, security systems, and regular updates can reduce this risk.

7. Quality Attributes

Quality attributes are characteristics that determine whether a product is good or not. Therefore, the key attributes of a hospital management system are listed below:

- **Accuracy:** The application should ensure that all data are stored and retrieved correctly. This will prevent financial losses and support the delivery of high-quality care .
- **Timeliness:** Information should be processed and delivered on time. For instance, a laboratory result should be available to a doctor as soon as possible to establish the diagnosis. In cases when a condition is detected timely, the patient has more chances for a full recovery.
- **User Satisfaction and Usability:** Clinicians should consider the HMS a helpful tool, not a difficult task. A high level of usability will ensure that medical workers will have less stress due to technology.

8. Artificial Intelligence in Hospital Management

Artificial intelligence (AI) is increasingly utilized in the health care sector. Researchers predict that for now, AI will not replace health care professionals, but it will optimize their work soon. The application of artificial intelligence in hospital management systems offers a variety of benefits to the users:

- **Health care analytics:** AI studies old patient records and hospital data. It looks for patterns and useful information. This helps doctors make better decisions and helps hospital managers improve hospital services and quality of care. For example: AI can check hospital data and show which diseases are increasing or which treatments are working better.
- **Precision medicine:** AI looks at a patient's medical history, chronic diseases, lifestyle, and genes. Based on this information, it can help doctors choose a treatment that is more suitable for that patient. For example: Two patients may have the same disease, but AI can help identify which treatment may be more appropriate for each patient based on their individual records.
- **Predicting diseases and illnesses:** AI uses a patient's previous health data and risk factors to predict whether they may develop a disease in the future. For example: AI may identify that a patient has a high risk of developing diabetes, allowing doctors to act early and provide timely treatment or prevention.
- **Interpret tests and diagnosed diseases:** AI can analyze medical tests such as X-rays, MRI scans, and CT scans very quickly. It can identify signs of problems such as cancerous tissues, fractures, and other diseases. This helps doctors make or confirm a diagnosis more efficiently, for example: AI can examine an X-ray and point out a possible bone fracture, which helps the doctor diagnose the problem.
- **Operational Optimization:** By analyzing the previous weeks/months records, AI predicts how many patients will be admitted to the hospital tomorrow/next week. For example, if every winter people get sick more often, the AI will notify the manager a month before that there will be about 50 patients with flu. Accordingly, the manager will prepare 50 beds and call more doctors/nurses for duty. He/she will also buy additional medicines and injections to ensure that no patients will wait for them.

9. Risks of Incorrect AI Usage

Artificial intelligence uses big data to identify diseases and determine the risk exposure of patients.

- **Diagnostic Errors and Lack of Accountability:** AI systems have the potential to commit errors in diagnosing and treating disease, which can result in harm to patients. There are also challenges concerning the accountability of AI for the errors that can happen during the implementation of the system. The issue of who is accountable for the errors made by AI is contentious, with the clinician's responsibility questioned and the developer's accountability rejected due to the lack of control over the process.
- **Availability and Access to Data:** The availability of data from different sources is key to the training of AI systems and algorithms utilized in health care. The issue of data accessibility in health care creates barriers to the development of successful models. Availability of data

in health care can be an issue as it is distributed over networks and institutions. The continuous availability of data for the enhancement of ML systems can also be an issue due to the unwillingness of various entities to share information

- **Transparency Issues:** The use of intelligent machines in the decision-making of health care raises concerns about accountability, transparency, permission, and privacy. Interpreting and understanding AI systems, such as the deep learning algorithms used in image analysis, can be challenging. Physicians who have no comprehension of the inner workings of AI models may have difficulty communicating the medical treatment process to patients
- **Risks of Data Sharing:** The rapid emergence of new technologies in health care has sparked skepticism due to the risks associated with data sharing. There is a need for public norms to ensure data governance and openness and improve patient understanding of how and why data are used. Patients may perceive this as a violation of their privacy, especially if the findings are made public to third parties.